

Survey of General End-to-End Autonomous Driving: A Unified Perspective

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Abstract—The quest for autonomous driving converges on the fundamental challenge of creating an integrated system that seamlessly maps raw sensory inputs directly to driving decisions. This ambition has catalyzed a paradigm shift from classic modular pipelines toward data-driven, end-to-end (E2E) approaches, aiming to overcome inherent limitations like information loss and error accumulation. In recent years, the E2E landscape has rapidly diversified into three prominent yet seemingly disparate paradigms. These include vision-centric methods that jointly optimize specialized modules, VLM-centric approaches that harness the world knowledge of large foundation models, and hybrid models that synergize the complementary strengths of the former two. While architecturally distinct, we posit that these are not divergent paths but rather different dialects of a common language. At their core, all three paradigms strive to solve the same *General End-to-End (GE2E)* problem. However, the literature lacks a unified survey that systematically analyzes and contrasts them from this holistic viewpoint. This paper bridges that critical gap. Through a comprehensive review of over 200 papers, we first dissect the architectures, design philosophies, and learning strategies that characterize each paradigm. Subsequently, we present a consolidated overview of mainstream evaluation benchmarks and a comparative performance analysis. By synthesizing their commonalities, we distill the field’s core challenges and chart a course for future research. Our work aims to provide a foundational resource and a guiding compass for researchers navigating the dynamic landscape of end-to-end autonomous driving. Our project page is available at <https://github.com/AutoLab-SAI-SJTU/GE2EAD>.

Index Terms—Autonomous Driving, End-to-End Model, Vision-Language Model, Policy Learning



1 INTRODUCTION

AUTONOMOUS driving (AD) represents a complex challenge that necessitates precise perception and contextual modeling of intricate scenarios to achieve real-time, safe vehicle control. The conventional autonomous driving system decomposes this task into several distinct functional modules, *e.g.*, perception, prediction, and planning, each trained in isolation and subsequently integrated into a unified framework. While this approach offers advantages in terms of debuggability and interpretability, it is fundamentally constrained by inherent limitations including information loss and error propagation, resulting in suboptimal system performance, particularly in complex and rare driving scenarios. To address these challenges, the field is increasingly adopting more robust, data-driven end-to-end autonomous driving frameworks, facilitated by the burgeoning availability of large-scale driving datasets.

End-to-end (E2E) systems for autonomous driving are defined as architectures that directly map raw sensor inputs to vehicle trajectories or control actions. Early conventional E2E approaches like P3 [1] (Fig. 1(a)-(b)) manually integrated one or more sub-tasks into a unified network for joint training and inference. This integration mitigates information loss and error propagation, enabling more robust planning and more efficient model architectures. Although

such paradigms constitute integrated systems, their internal modules exhibit distinct task-specific specialization.

With the rapid advancement of vision-language models’ (VLMs) multimodal understanding and reasoning capabilities, a novel VLM-centric E2E paradigm (Fig. 1 (c)) has emerged. Distinct from conventional E2E systems, this paradigm centers on a general-purpose, knowledge-rich foundation VLMs that internalizes extensive world knowledge. By projecting multimodal data into a linguistic space, these frameworks leverage language generation or external action decoders to enable language models to perform low-level vehicle control, marking a transformative shift in autonomous driving architectures.

Our work reveals significant distinctions and intrinsic connections between two prevailing paradigms. Conventional E2E frameworks ground their decisions in explicit and detailed 3D scene representations, such as occupancy grids or Bird’s-Eye-View (BEV) object detections. This approach provides a structured understanding of the environment, leading to highly reliable motion planning within well-defined scenarios. However, this reliance on pre-defined geometric priors often limits their generalization capabilities when encountering unseen or long-tail driving situations. In contrast, VLM-centric E2E harnesses the extensive world knowledge embedded within foundation models, which are pre-trained on vast, web-scale datasets. This enables them to exhibit remarkable generalization capabilities and robustly handle ambiguous, long-tail scenarios that often challenge conventional models, despite notable limitations in trajectory generation precision and computational efficiency. Consequently, the complementary

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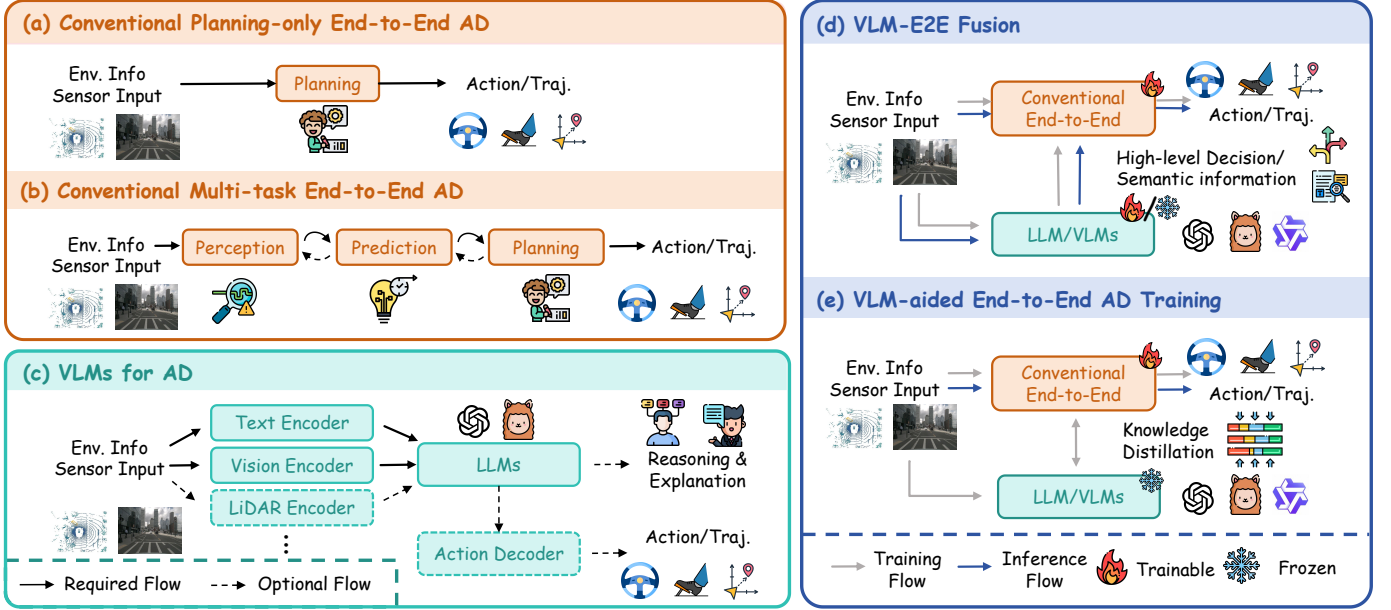


Fig. 1: Comparison of different end-to-end autonomous driving paradigms. (a) **Conventional Planning-only E2E:** Direct input-to-action mapping, which lacks interpretability. (b) **Conventional Multi-task E2E:** Leveraging multiple tasks to improve robustness. (c) **VLMs for AD:** Employing VLM reasoning for more interpretable and robust driving behaviors. (d) **VLM-E2E Fusion:** A VLM serves as a high-level reasoner guiding a low-level E2E executor. (e) **VLM-aided E2E Training:** Distilling VLM knowledge into an E2E model during training to efficiently boost performance.

strengths of these paradigms provide both motivation and feasibility for the emergence of hybrid architectures.

Hybrid paradigms emerge as a powerful synthesis, aiming to integrate the strengths of both conventional and VLM-centric E2E frameworks. These approaches generally manifest in two distinct strategies. The first strategy involves an online collaboration, where a VLM acts as a high-level reasoning engine during inference. It provides semantic guidance or strategic planning advice to the conventional E2E model, which remains responsible for fine-grained motion execution, akin to a "thinking slow, acting fast" architecture (Fig. 1 (d)). The second strategy employs an offline knowledge transfer through distillation. In this setup, the VLM serves as a teacher model exclusively during the training phase, injecting its rich semantic understanding into the more compact conventional E2E model. Consequently, the VLM is discarded during inference, leaving an efficient yet semantically-aware planning model (Fig. 1 (e)). By strategically weaving together the VLM's generalization prowess with the E2E model's safety and efficiency, these hybrid methods have pushed the state-of-the-art (SOTA) across planning accuracy, interpretability, and safety assurance.

Despite the differences in structure, we observe significant similarities across the three aforementioned paradigms in terms of learning strategies, dataset selection, and prevailing challenges. For learning strategies, the majority of methods within these paradigms employ imitation learning, where the model learns from expert demonstrations. Concurrently, similar advanced learning strategies are being explored to address the inherent limitations of imitation learning, such as overfitting, causal confusion, and poor generalization. In terms of dataset selection, the different paradigms are benchmarked on largely consistent datasets,

which facilitates a holistic performance comparison across them. For research challenges, the long-tail nature of data, as well as the interpretability, safety, and real-time performance of autonomous driving systems, constitute common problems that all paradigms are committed to addressing.

Building on this, this paper conducts a systematic review of the three end-to-end autonomous driving paradigms from a unified perspective. First, we review the representative works within each paradigm, introducing their respective architectures, functional designs, and learning strategies. Subsequently, we review the primary datasets, including datasets without and with language annotations, and conduct a performance analysis on key benchmarks. Following this, we analyze the persistent challenges confronting end-to-end autonomous driving from a unified standpoint. Finally, based on the preceding analysis, we propose potential future research directions, aiming to guide researchers in the field.

1.1 Comparison with Other Surveys

Although existing surveys partly reviewed related topics, such as conventional end-to-end driving [2, 3, 4], the application of VLMs [5, 6, 7, 8], they treat these paradigms as separate and disconnected research threads. Consequently, a systematic analysis that examines the relationships and distinctions between them through this unifying lens is critically absent, leading to fragmented coverage of the emerging hybrid paradigms. This paper introduces the concept of General End-to-End (GE2E) driving. We define GE2E as any paradigm where raw sensor inputs are processed into planning trajectories or control actions via a holistic model, regardless of whether the architecture incorporates VLMs or not. Under this framework, we posit that these seemingly

disparate approaches are, in fact, distinct manifestations of a single, unified problem. To our knowledge, this work is the first to offer a cohesive synthesis by framing the field through the GE2E perspective, systematically reviewing conventional, VLM-centric, and hybrid methods with a focus on the latest research progress.

1.2 Contributions of the Survey

- This survey systematically analyzes general E2E autonomous driving paradigms and provides an in-depth analysis of their respective architectures, functional designs, and learning strategies. For visualization, we present a comprehensive mind map (Fig. 2) and a detailed architectural comparison table (Tab. 1).
- We compile and review key datasets used in general E2E autonomous driving, encompassing both conventional benchmarks and emerging VL datasets, summarized in a clear tabular format (Tab. 2). Furthermore, a horizontal comparative analysis of open-loop and closed-loop performance across different paradigms is conducted on mainstream benchmarks, as illustrated in Fig. 7, offering empirical evidence to evaluate the current state of the art.
- We move beyond the discussion of individual paradigms to provide a unified synthesis of the common challenges facing the entire field. Building on this foundation, the survey outlines promising future research avenues, providing strategic insights and a potential roadmap for the field’s continued progression.

2 CONVENTIONAL END-TO-END AUTONOMOUS DRIVING

In the early stages of autonomous driving development, the prevalent approach adopts a modular architecture rather than an end-to-end methodology. This design often decomposes the system into three core sub-tasks of perception, prediction, and planning, with each sub-task handled by an independently trained specialized module. Within the integrated system, these modules processed data sequentially, with the output of one module serving as the input for the subsequent module. While this paradigm provides notable advantages in interpretability and debugging simplicity, the inherent isolation and independent training of modules lead to information loss, error accumulation, and feature misalignment across the pipeline due to conflicting optimization objectives between modules [9, 10].

To address the limitations of modular approaches, conventional end-to-end autonomous driving frameworks like P3 series [1, 11, 12], Transfuser [13], UniAD [9] integrate perception, prediction, and planning into a unified optimization process focused on the ultimate planning objective. Such model architecture design has three main advantages: (a) *Less Information Loss*. By learning an intermediate representation directly optimized for planning, the system minimizes information loss inherent in sequential task execution. (b) *Less Error Propagation*. Sub-modules are trained to perform robust planning despite imperfect prior perception and prediction results, further alleviating the issue of error propagation. (c) *Effective Model Architecture*. A

highly integrated model architecture facilitates deployment and enhances computational efficiency, with the system capabilities scaling directly with increasing training resources.

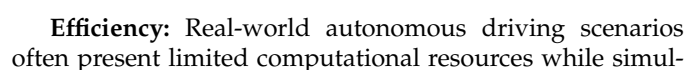
It is worth noting that conventional end-to-end autonomous driving does not necessarily explicitly define separate perception and prediction sub-modules. Consequently, based on the presence or absence of explicit sub-module design, we categorize end-to-end model architectures into *Planning-Only End-to-End Methods* (Sec. 2.1) and *Multi-Task End-to-End Methods* (Sec. 2.2). The following sections detail the primary methods for each paradigm, along with their respective advantages and limitations. Finally, we will further talk about the innovations of the learning strategies (Sec. 2.3) in the conventional E2E paradigm, which transcend supervised imitation learning and data constraints to achieve better efficiency and performance.

2.1 Planning-Only End-to-End Methods

Limited by the paucity of computational resources and data, the earliest end-to-end autonomous driving models are architecturally simplified, often comprising a singular planning module, as illustrated in Fig. 1 (a). For instance, Alvin [14], a pioneering work in end-to-end autonomous driving, employs a simple three-layer neural network to map inputs from cameras and a laser range finder directly to steering outputs. NVIDIA later advances an end-to-end CNN [15], directly converting raw images into steering commands. Following these seminal works, subsequent methods primarily focus on optimizations in three aspects: 1) *multimodal fusion* for 3D understanding enhancement, 2) *generative modeling* that reflects multimodal characteristics, and 3) *efficiency* for practical on-vehicle implementation.

Multi-modal Fusion: The aforementioned early methods either solely use camera images as model input or do not adequately fuse different modalities. Consequently, these approaches fail to extract sufficient 3D scene spatial information, thereby limiting their performance. A pivotal turning point is the introduction of TransFuser [13], which introduces an attention-based multimodal fusion Transformer to effectively combine image and LiDAR inputs. Similarly, MMFN [16] also utilizes a transformer-based framework, extending the fusion process to integrate inputs from cameras, LiDAR, HD Maps, and radar. These methods provide informative scene representations for the planning task, making the planning results safer and more reliable.

Generative Modeling: Despite these advancements, the above methods exhibit certain inherent limitations. That is, they all employ a regression-based approach to fit ground-truth driving behaviors for specific scenarios as closely as possible. However, under identical scene conditions, the ego-vehicle may have multiple candidate behaviors it can adopt. This phenomenon is known as the multimodality of trajectory prediction. For instance, when a vehicle stops ahead, the agent can choose to stop or change lanes safely to achieve more efficient driving. Direct regression prediction in such scenarios could lead to an averaged, potentially unsafe, behavior [17]. Consequently, some recent approaches have begun exploring generative models (e.g., diffusion models [18, 19, 20]) for probabilistic planning. These models can inherently model the distribution of trajectory



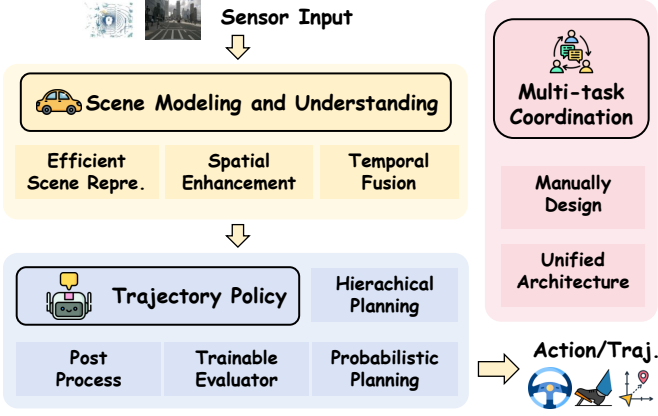


Fig. 3: Main innovations of the architecture of multi-task E2E autonomous driving, including Scene Modeling and Understanding (Sec. 2.2.1), Multi-task Coordination (Sec. 2.2.2), Trajectory Policy (Sec. 2.2.3).

taneously imposing stringent real-time requirements due to safety imperatives. To reduce the computational cost of LiDAR data, Fast-LiDARNet [23] implements efficient hardware optimizations, achieving efficient and robust end-to-end control. In addition, the aforementioned methods, such as the introduction of additional parameters in fusion modules and the high latency associated with generative modeling, incur overheads that hinder practical deployment. To address this challenge, more recent efforts like EfficientFuser [24] reduce the required parameters and computational load through the lightweight EfficientViT network, while DRAMA [25], GMF-Drive [26], and MambaFusion [27] adopt Mamba-based architectures to achieve lower computational complexity. These methods enhance the feasibility of on-board deployment by reducing computational demands while simultaneously preserving performance.

However, due to the inherent simplicity of planner-only model architectures, they suffer from two intrinsic limitations: (1) *Lack of Interpretability*. The model, designed to map directly from raw sensor input to final trajectory generation without any intermediate outputs, is considered a black box. This inherent opaqueness leads to a lack of both safety assurance and interpretability, making it challenging to gain user trust. Specifically, when such a model produces unsafe driving behavior, the absence of a discernible decision-making process hinders the analysis of the root causes of failure. (2) *Sparsity of Supervision*. This simplified architecture does not leverage 3D ground truth data of perception and prediction, instead relying solely on planning trajectory supervision. On one hand, the model’s 3D understanding capabilities are insufficiently trained, making it challenging for the model to comprehend complex driving scenarios. On the other hand, by relying solely on sparse planning signals to imitate driver trajectories, the model struggles to learn the intrinsic logic of the driving process, which can lead to unsafe driving behaviors.

2.2 Multi-Task End-to-End Methods

As the preceding discussion highlighted, due to the opacity and lack of 3D prior information in planning-only end-to-end model architectures, current mainstream end-to-end

approaches are gradually shifting towards modular multi-task end-to-end architectures, as illustrated in Fig. 1 (b). Some early methods [1, 9, 11, 12] introduce perception and prediction tasks, using these as additional supervision for intermediate representations, thereby forming a unified end-to-end architecture that fosters collaboration among tasks. Subsequent work focuses on optimization across three key aspects (see Fig. 3): *Scene Modeling and Understanding* (Sec. 2.2.1), *Multi-tasks Coordination* (Sec. 2.2.2) and *Trajectory Generation Policy* (Sec. 2.2.3), to enhance crucial autonomous driving requirements of robustness and interpretability.

2.2.1 Scene Modeling And Understanding

Scene modeling and understanding are fundamental tasks in autonomous driving. In end-to-end multi-task frameworks, beyond performing basic perception tasks like detecting agents and maps, it is also required to provide a scene representation that incorporates global and temporal information to assist downstream prediction and planning tasks. Then, how to better model the relationships between dynamic agents and static maps, and how to better achieve temporal fusion and model the flow of scenes in complex driving scenarios, pose significant challenges.

Efficient Scene Representations: Early modular end-to-end methods [1, 9, 11, 12] rely on dense rasterized scene representation (e.g., BEV map). For instance, UniAD [9] leverages BEVFormer [28] to implicitly model the mapping relationship between 2D BEV space and multi-view image features, thereby obtaining a dense BEV representation that serves as the environmental representation for downstream tasks. Then, the model utilizes the predicted occupancy for post-processing forecasted trajectories. However, such an approach is computationally intensive and misses the instance-level structure information [29]. Therefore, some research has begun to explore more efficient instance-level scene representations. For instance, VAD [29] enhances planning safety by incorporating vectorized planning constraints rather than processing dense occupancy representations. Furthermore, for autonomous driving perception tasks that solely require attention to specific driving elements, a BEV of the entire scene often contains substantial redundant information. Consequently, sparse methods such as SparseAD [30] and SparseDrive [31] opt to directly model the BEV features of object instances rather than dense, scene-wide BEV features. Specifically, the model acquires the visual features of targets directly from multiple images via object instance queries, and these features are then utilized to predict the targets’ 3D information. This approach circumvents the need for dense BEV feature modeling, thereby reducing computational load while concurrently focusing on instance-level information and enhancing the driving performance. In addition, GaussianAD [32] introduces another direction by leveraging 3D semantic Gaussians to describe the scene extensively yet sparsely, which has demonstrated its effectiveness across various tasks.

Spatial Understanding and Reasoning: To achieve more robust autonomous driving performance, driving policies require not only compact scene representations but also a deep spatial understanding and reasoning about the scene. Therefore, some methods introduce prior knowledge about agent-map and agent-agent relationships within the scene to

enhance comprehension of the interactions among different scene elements. For example, GraphAD [33] employs graph models to describe complex interactions among agents and maps within traffic scenarios, while DualAD [34] decouples dynamic agents and static scene elements modeling. Other methods, such as FusionAD [35], enhance the spatial information of BEV features by fusing LiDAR and camera inputs, thereby further improving the model’s spatial understanding capabilities. Given the inherent difficulties in modeling long-tail scenarios and the imperfections of perception results, to better comprehend the scene, UncAD [36] performs uncertainty modeling of the scene and leverages this uncertainty to guide motion prediction and planning modules to produce multimodal trajectories, which enhances the model’s generalization ability and robustness.

Temporal Fusion and Reasoning: Beyond enhancing the policy’s spatial understanding of the scene, other methods are dedicated to enhancing the model’s comprehension of scene dynamics and temporal characteristics, thereby improving the model’s performance on perception and prediction tasks. Conventional approaches [28, 37, 38, 39] typically rely on perception-level temporal fusion, iteratively aligning and merging features from adjacent timesteps. To capture the scenic dynamics more explicitly in multi-task E2E pipelines, some novel methods, such as ReasonNet [40], store historical information in a long-short term memory bank and employ a Transformer-based global reasoning module for enhanced scene comprehension. BridgeAD [41], on the other hand, segments planning trajectories by timestep, aligning trajectory prediction information from historical frames with the current frame’s timestep, thereby enhancing present-moment perception, prediction, and planning tasks. More recently, some methods argue that when prediction and planning tasks are performed serially, the influence of the ego-vehicle’s trajectory on other agents’ behavior is inadequately considered, which is unreasonable. GenAD [42] maps scene representation to a structural latent trajectory space and employs a future trajectory generator to produce future trajectories, thereby simultaneously completing motion prediction and planning. DiffAD [43] redefines autonomous driving as a conditional image generation task, generating BEV images conditioned on BEV features to unify various driving objectives and better model the temporal evolution of the environment.

2.2.2 Multi-task Coordination

The potential for different tasks to collaborate and achieve positive transfer is a significant advantage of end-to-end autonomous driving. However, early multi-task E2E frameworks often execute perception, prediction, and planning tasks in a strict sequential manner. Some research [44] argues that this sequential design limits inter-task transfer, leading to suboptimal performance. Subsequent methods achieve enhanced task collaboration by adopting manually designed task coordination strategies or by integrating all tasks within a unified framework to enable deep interaction, allowing the model to autonomously learn task cooperation.

Manually Designed Coordination: Some approaches aggregate or decouple subtasks based on their inherent similarities or differences, thereby facilitating optimal positive transfer among them. For instance, PPAD [45] models

dynamic environments by alternating between prediction and planning tasks, reflecting the inherent characteristic that other agents’ movements are influenced by the ego-vehicle’s planning. TTOG [46] unifies the motion and planning tasks, enabling the planning task to benefit from motion data, which significantly improves its performance and generalizability. Recognizing the inherent gap between scene perception and motion prediction tasks, DriveAdapter [47] decouples the perceptual learning of the student model from the planning knowledge of the teacher model by introducing an adapter module, thereby avoiding the causal confusion problem inherent in traditional behavioral cloning methods. Similarly, DMAD [48] separates semantic learning from motion learning, decoupling the gradient updates for these two tasks to eliminate negative transfer and exploiting inter-task correlations to promote positive transfer.

Unified Multi-task Architecture: Some studies contend that human-designed task sequences restrict the model’s ability to autonomously learn inter-task collaboration and are less conducive to scaling up. Therefore, methods like DriveTransformer [44] and HiP-AD [49] directly interact all subtask queries within a single module for deep collaborative processing. Simultaneously, it sparsely models the environment through direct interaction between each subtask’s query and the raw camera input, while also deeply fusing with historical queries to capture temporal information. Then, these queries are fed into their respective task-specific decoder heads. This architecture fosters deep integration and interaction among sub-tasks, offering a potential avenue for scaling up end-to-end systems.

2.2.3 Trajectory Planning Policy

This section mainly introduces the methods of generating safe and stable trajectories based on scene analysis and understanding. Simple imitation learning is often considered to merely mimic driving behaviors without understanding driving logic, which can lead to models that do not guarantee safe, comfortable, or traffic-rule-compliant actions. To address this challenge, subsequent methods utilize post-processing, trainable scorers, generative modeling, and hierarchical planning methods to ensure more reliable driving behaviors.

Post-processing Optimization: Some existing methods mitigate this issue by designing cost functions for post-processing. For instance, UniAD [9] optimizes trajectories further using Newton’s method combined with predicted occupancy. Such a design allows the model to leverage 3D information for adaptive adjustments. However, such approaches render the end-to-end pipeline less streamlined, and imperfect perception results can potentially degrade optimization outcomes. Furthermore, the performance of this post-processing method cannot be enhanced through learning-based approaches, thereby restricting its scalability. Therefore, more recent works [50, 51] adopt differentiable optimization processes, which enable the post-processing to continuously improve its performance with training.

Trainable Trajectory Evaluator: Beyond the differentiable optimization, some works attempt to explore a new paradigm that selects an optimal trajectory from the pre-defined trajectory vocabulary (obtained by clustering ground-truth trajectories) via a trainable evaluation selector

in an end-to-end training manner. This ensures a streamlined model architecture, and as the training process goes on, the robustness and performance of the evaluators can be enhanced. For instance, Hydra-MDP [52, 53] features a multi-head decoder to learn diverse trajectory candidates tailored to various evaluation metrics, winning first place in the 2024 NAVSIM challenge [54]. WoTE [55] and World4Drive [56] leverage a World model to predict future context states for trajectory evaluation, achieving SOTA performance on the NAVSIM benchmark.

Probabilistic Trajectory Planning: Regarding the issue of the planning task’s non-deterministic nature mentioned in Sec. 2.1, direct regression methods may predict suboptimal intermediate values, which can also lead to safety concerns. VADv2 [17] tries to solve this problem by sampling a single action from the predicted probabilistic action distribution, improving closed-loop performance on CARLA Town05 benchmark. Subsequently, several methods [31, 57] have also begun to predict multimodal trajectories and apply additional selection strategies based on the resulting probabilistic distributions.

As diffusion policies emerge as powerful techniques in robot policy learning, some works have begun to employ the strategy in autonomous driving. A notable challenge, however, is that autonomous driving tasks demand significantly higher real-time performance than robot policies. To address this challenge, DiffusionDrive [58] proposes a novel truncated diffusion strategy, enabling the model to learn denoising from an anchored Gaussian distribution. This approach achieves high performance while operating at 45 FPS on an NVIDIA 4090, mitigating the high latency and making diffusion strategies feasible for autonomous driving. Consistency [59] jointly samples the future trajectories of the self-vehicle and surrounding traffic participants through the consistency model, and combines the navigation objectives of the self-vehicle to generate high-quality interactive trajectories. DiffE2E [60] combines diffusion models with explicit supervision and employs a hybrid decoding and collaborative training mechanism to address insufficient generalization capabilities.

Hierarchical Planning: Some studies argue that the one-shot trajectory inference paradigm (generating all waypoints in a time) fails to capture the dynamic evolution of the environment. To address this, KEMP [61] achieves efficiency and accuracy in long-term trajectory prediction through hierarchical keyframe forecasting. In contrast, ThinkTwice [62] enhances driving decision-making capabilities by stacking multiple decoder modules to iteratively refine prediction outcomes using spatio-temporal priors. To emulate the hierarchical cognition mechanisms of human drivers, CogAD [63] adopts a coarse-to-fine method, which first implements intent-level planning and subsequently trajectory-level planning. In contrast, GEMINUS [64] leverages the MoE framework to achieve scene-adaptive planning by an uncertainty router. Other methods like ARTEMIS [65] implement autoregressive methods to serially generate trajectory points, ensuring temporal dependency while significantly boosting trajectory generation robustness.

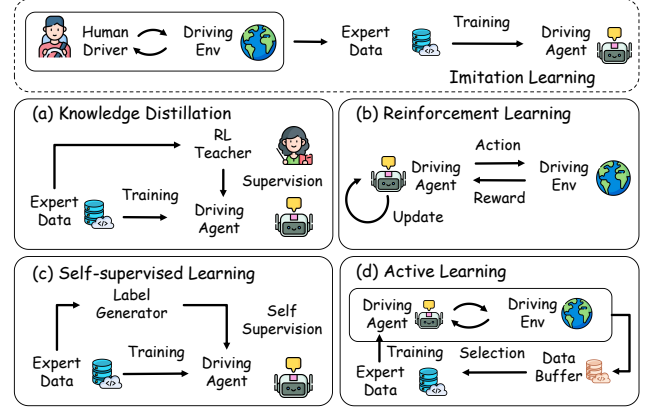


Fig. 4: Learning strategies in conventional E2E autonomous driving, including knowledge distillation (Sec. 2.3.1), reinforcement learning (Sec. 2.3.2), self-supervised learning (Sec. 2.3.3) and active learning (Sec. 2.3.4).

2.3 Learning Strategy

As shown in Fig. 4, most methods employ imitation learning to learn driving behavior from expert demonstrations. For instance, CIL [67] uses conditional imitation learning to achieve controlled autonomous driving in complex scenarios by setting up an independent network for each steering, while Multitask-with-attention [91] combines the conditional imitation learning framework and multi-task learning to design an attention-aware network to enhance the generalization ability of the model. However, such learning paradigms often exhibit poor generalization when handling long-tail scenarios. This section focuses on the optimization of learning strategies and how they can push through the limitations of data to achieve a more robust driving system.

2.3.1 Knowledge Distillation

The efficacy of imitation learning for autonomous driving is often constrained by the sparse supervisory signals inherent in driver actions, a limitation that persists even with large-scale datasets [92]. To surmount this challenge, a prominent line of research employs knowledge distillation, wherein a privileged teacher model provides dense, intermediate supervision to a student that operates on raw sensory inputs.

This paradigm is pioneered by methods utilizing agents with access to privileged information. LBC [68], for example, trains an agent with access to ground-truth environmental states to distill its operational knowledge into a non-privileged, vision-based student. Building upon a similar principle, Roach [93] employs a privileged agent trained via reinforcement learning to generate high-quality, dense supervision, thereby enhancing the learning efficacy for the end-to-end student model. Subsequent research has evolved to distill more complex and structured intermediate representations. IVMP [94] supervises its student on the tasks of semantic maps prediction and trajectory planning with optical flow information to improve network performance. Extending this approach, DistillDrive [95] utilizes an RL-based multimodal planning model as the teacher, guiding student models to learn multimodal motion features through distillation to enhance the planning performance.

TABLE 1: An architectural overview of various end-to-end autonomous driving models.

- **Input:** Camera, LiDAR, Ego Status, Driving Command.
- **Backbone:** VE: Vision Encoder (Res: ResNet, Vov: VovNetV2, Effi: EfficientNet-B4, Point: PointPillars), LD: Language Decoder.
- **Temporal Fusion:** denotes the method for processing sensor inputs across multiple timesteps.
- **Intermediate Tasks:** refer to the sub-tasks decomposed from the end-to-end planning process. *Perception* (**M** Mapping, **D** Detection, **T** Tracking, **SU** Scene Understanding, **SD** Scene Description), *Prediction* (**2D** 2D Perception, **O** Occupancy Prediction, **Mo** Motion Prediction, **W** Latent World Model, **IP** Intention Prediction). The *Planning* task is mainly classified according to the methods: *Deterministic* (**R** Regression, **CS** Candidates Selection, **DO** Deterministic Optimization, **AR** Autoregressive, **TA** Text Autoregressive), and *Probabilistic* (**S** Probabilistic Sampling, **PS** Probabilistic Selection, **PO** Probabilistic Optimization, **PG** Probabilistic Generative).
- **Output:** Trajectory, Control Signal, Meta Action, CoT, Text Answers.
- **Learning Strategy:** IL: Imitation Learning, RL: Reinforcement Learning, KD: Knowledge Distillation, SSL: Self-supervised Learning.

Category	Input				Backbone		Scene	Temporal	Intermediate Tasks			Interaction	Output	Learn. Stra.	Representative Work
					VE	LD	Representation	Fusion	Perc.	Pred.	Plan.	between subtasks			
Planning-only E2E	✓	✓	-	-	MLP	-	-	-	-	-	R	-		IL	ALVINN [14]
	✓	-	-	-	CNN	-	-	-	-	-	R	-		IL	CNN-E2E [15]
	✓	-	-	-	CNN	-	-	-	-	-	R	-		RL	Drive in a Day [66]
	✓	-	✓	✓	CNN	-	-	-	-	-	R	-		IL	CIL [67]
	✓	-	✓	✓	Res.	-	-	-	-	-	R	-		KD	LBC [68]
	✓	-	✓	✓	Res.	-	-	-	-	-	R	-		IL	DARB [69]
	✓	-	✓	-	Res.	-	Implicit	Iterative	-	-	R	-		IL	NEAT [70]
	✓	✓	-	-	Res.	-	Implicit	-	-	-	R	-		IL	Transfuser [13]
Multi-task E2E	✓	✓	✓	✓	Mamba	-	Implicit	-	-	-	R	-		IL	DRAMA [25]
	-	✓	✓	-	Res.	-	Occ Voxel	-	-	-	O	Sequential		IL	P3 [1]
	✓	✓	✓	✓	Res. & Point.	-	BEV	Memory bank	-	-	O	Parallel pred. plan.		IL	ReasonNet [40]
	✓	-	✓	✓	Effi.	-	BEV	Iterative	M	O	DO	Sequential		IL	ST-P3 [11]
	✓	-	-	✓	Res.	-	BEV	Iterative	D T M	Mo O	DO	Sequential		IL	UniAD [9]
	✓	-	✓	✓	Res.	-	Sparse	-	T M	Mo	R	Sequential		IL	VAD [29]
	✓	-	✓	✓	Res.	-	Sparse	-	T M	Mo	S	Sequential		IL	VADv2 [17]
	✓	-	✓	✓	Res.	-	Scene Graph	Multi-step Aggregation	T M	Mo O	PO	Sequential		IL	GraphAD [33]
	✓	-	✓	-	Res.	-	Sparse	-	O M	Mo	PG	Sequential		IL	GenAD [42]
	✓	✓	✓	✓	Res.	-	Sparse	Memory bank	D T M	Mo	PO	Parallel pred. plan.		IL	SparseAD [30], SparseDrive [31]
	✓	-	✓	-	Vov.	-	Sparse & BEV	Iterative	D T M	Mo	PO	Dynamic-static		IL	DualAD [34]
	✓	✓	✓	✓	Res.	-	Implicit	Iterative	D	-	CS	Sequential		KD	Hydra-MDP [52]
	✓	-	-	✓	Res.	-	BEV	Iterative	D	W	R	Sequential		SSL	UAD [71]
	✓	-	✓	✓	Res.	-	BEV	Iterative	D M	Mo	R	Iterative Pre. Plan.		IL	PPAD [45]
	✓	-	✓	✓	Res.	-	BEV	Iterative	D M	Mo	PS	Sequential		IL	UncAD [36]
	✓	-	✓	-	Res.	-	Sparse	Memory Bank	O M	Mo	PO	Unified		IL	DriveTransformer [44], HiP-AD [49]
	✓	✓	✓	-	Res.	-	BEV	-	-	-	PO	-		IL	GoalFlow [21]
	✓	-	-	✓	Res.	-	Sparse	Memory bank	D T M	Mo	DO	Memory Interaction		IL	BridgeAD [41]
	✓	-	-	-	Res.	-	Sparse	Memory Bank	D T M	Mo	PS	Sequential		IL	MomAD [57]
VLM-Centric E2E	✓	-	-	✓	Res.	-	Latent-WM	Iterative	2D	W	PS	Sequential		SSL	World4Drive [56]
	✓	-	✓	-	ViT	LLaMA2	Visual Token	Video encoder	SU	-	TA	-		PT & SFT	DriveGPT4 [72]
	-	-	✓	-	-	LLaMA-7B	Vector	-	SD	-	TA	-		SFT	Driving with LLMs [73]
	✓	✓	-	✓	ViT-G. & GD-MAE	LLaMA-7B	Visual Token	Iterative	-	-	TA	-		SFT	DriveMLM [74]
	✓	✓	-	✓	Res. & PointNet	Vicuna-v1.5-7B	BEV	-	-	-	R	-		SFT	LMDrive [75]
	✓	-	-	-	ViT-G	Vicuna-7B	Video with prior	Aggregation	SD	Mo	-	-		SFT	Reason2Drive [76]
	✓	-	-	-	ViT	Qwen1.5	Visual Token	-	SU	-	R	-		SFT	LeapAD [77]
	✓	-	✓	-	Res.	GPT3.5-turbo	Text info.	-	D M	-	TA DO	-		SFT	AgentDriver [78]
	✓	-	-	-	ViT	FlanT5	Visual Token	-	D	IP	TA	GVQA		SFT	DriveLM [79]
	✓	-	✓	-	ViT	Vicuna 1.5 7B	Visual Token	Video encoder	-	-	TA	-		PT & SFT	RAG-Driver
	✓	-	✓	✓	-	Gemini1.0	Visual Token	-	SU	-	TA	-		SFT	EMMA [80]
	✓	-	✓	✓	ViT	Vicuna-v1.5-7b	Visual Token	iterative	SD D	IP	R	-		PT & SFT	Senna [81]
	✓	-	-	✓	EVA02	Vicuna-13B	Perception Query	-	D	-	TA	-		SFT	OmniDrive [82]
	✓	-	✓	✓	ViT-G	PaLI3-5B	Sparse Volume	Aggregation	-	-	TA	-		SFT	S4-Driver [83]
	✓	-	✓	✓	ViT	Qwen2.5	Visual Token	-	SD D	-	TA	-		SFT & RL	AutoVLA [84]
	✓	-	✓	-	ViT	Qwen-0.5B	Visual Token	-	-	-	R	-		SFT & AL	DriveGPT4v2 [85]
Hybrid E2E	✓	-	✓	-	ViT	Paligenma-3b	Visual Token	-	-	-	PG	-		SFT	DriveMoE [86]
	✓	-	✓	✓	Res	Qwen 2.5-Instruct	Perception query	-	T M	Mo	TA	-		SFT	OpenDriveVLA [87]
	✓	-	✓	-	ViT	Qwen	Visual Token	Aggregation	SU	-	TA DO	-		SFT	DriveVLM [88]
Hybrid E2E	✓	-	-	✓	Res	Qwen	Sparse	-	M D	-	PO	-		SFT & RL	FasionAD [89]
	✓	-	✓	-	EVA02	LLaMA-7B	Visual Token	memory bank	M D	-	DO	-		PT & SFT	SOLVE [90]

2.3.2 Reinforcement Learning

Reinforcement learning (RL) provides an attractive alternative to imitation learning for autonomous driving. Through active exploration and trial-and-error, policies can discover rare long-tail scenarios beyond expert demonstrations. An early work of RL in autonomous driving is Drive in a Day [66], which introduced deep RL to driving by framing it as a Markov Decision Process. Exploration is crucial for improving RL. For more effective exploration, uncertainty has been used to guide policies toward valuable and unseen states, avoiding inefficient or unsafe trial-and-error. RDMF [96] employs Supervised Ensemble Deep Deterministic Policy Gradient, measuring model uncertainty by the variance of critic networks, and switches to a rule-based policy when thresholds are exceeded. SAPO-RM [97] applies probabilistic Control Barrier Functions to update constraints step by step, ensuring feasibility in high-risk scenarios. With the advent of modern modular architectures, the focus has shifted to integrating RL into closed-loop training systems that leverage world reconstruction to reduce risk and cost in real environments. RAD [98] introduces this paradigm by using 3D Gaussian Splatting to build digital replicas and applying the Proximal Policy Optimization for large-scale trial-and-error learning, which significantly reduces collisions. ReconDreamer-RL [99] extends this idea with video diffusion models to generate complex long-tail scenarios, enabling interactive RL optimization for better policy training. Building on this foundation, EvaDrive [100] combines closed-loop training with multi-objective adversarial optimization to reduce inefficient or risky trial-and-error.

2.3.3 Self-supervised Learning

Due to the scarcity and high cost of manually annotated data, self-supervised learning can effectively capture the underlying patterns within data, thereby reducing the model's reliance on large amounts of labeled data. PPGeo [101] pre-trains a visual encoder through a two-stage paradigm consisting of self-supervised geometric modeling and vehicle policy pre-training, providing effective feature representations for various downstream tasks with only a small amount of labeled fine-tuning data. LAW [102] employs a self-supervised learning approach based on a latent world model to optimize scene feature representations and future trajectory predictions. UAD [71] replaces perception and planning modules that depend on 3D annotations with 2D unsupervised tasks, achieving performance surpassing SOTA methods with zero 3D annotations while significantly improving efficiency. This offers a viable pathway for scaling end-to-end driving models using massive unlabeled datasets without the need for expensive annotations.

2.3.4 Active Learning

A critical problem in imitation learning is covariate shift, where policy deviations lead to compounding errors on unfamiliar states. The DAGger [103] algorithm, a widely adopted method inspired by active learning, mitigates this by iteratively collecting expert-labeled data from states the policy actually encounters. However, this naive aggregation can degrade performance over time. DARB [69] identifies the cause as redundant states and the long-tailed distribution of critical scenarios. It refines DAGger by introducing a

critical-state sampling mechanism, which prioritizes learning from the most informative data to reduce overfitting and improve robustness in safety-critical situations.

Beyond its primary role in mitigating covariate shift, active learning is also instrumental in enhancing data efficiency. ActiveAD [104] designs an active learning method that progressively annotates collected unlabeled data according to diversity and usefulness criteria, achieving full-dataset performance on 30% of nuScenes [105]. SEAD [106] selects initial data based on driving environmental information and incremental data based on BEV features, which further optimizes the data selection strategy, reduces annotation costs, and enhances performance. These findings underscore the critical importance of data quality over sheer quantity in autonomous driving.

2.3.5 Other Learning Strategy

Several studies leverage other advanced learning strategies to enhance the robustness and adaptability of end-to-end autonomous driving. For instance, MA2T [107] proposes a targeted adversarial training method to enhance model robustness against malicious attacks; Centaur [108] employs Test-Time Training (TTT) to enable real-time adaptation to uncertain scenarios during deployment; and CRCHFL [109] designs a hierarchical federated learning framework to address the challenge of distributed model training with limited communication resources. Collectively, these works advance autonomous driving learning frameworks towards safer and more reliable practical applications.

3 END-TO-END REASONING WITH VISION-LANGUAGE MODEL IN AUTONOMOUS DRIVING

While the conventional E2E paradigms detailed in the preceding chapter have demonstrated notable advancements, their inherent architectural constraints impose a fundamental ceiling on their generalization and reasoning capabilities. These limitations manifest primarily in two forms. First, their perceptual modules operate under a closed-world assumption, restricting scene interpretation to a finite, predefined taxonomy of objects and states. Second, the system architecture, typically a cascade of specialized sub-models for tasks such as perception and prediction, inhibits the formation of a truly holistic and unified scene representation. This mechanism, by its very nature, lacks the capacity for deep causal reasoning. This deficiency culminates in a significant cognitive gap, which we define as the system's intrinsic inability to achieve robust generalization and trustworthiness in long-tail scenarios. Such scenarios are precisely those that demand the application of external world knowledge and complex logical inference, capabilities that lie beyond the scope of conventional E2E architectures.

In response to this cognitive gap, a new paradigm has emerged centered on Vision-Language Models (VLMs), which seek to instantiate a "Cognitive Driving Agent" by grounding the vast world knowledge and inferential capabilities of Large Language Models (LLMs) within the spatiotemporal context of the driving scene, as depicted in Fig. 1 (c). Unlike the conventional E2E predecessors, these agents exhibit superior capabilities in comprehending complex, open-world situations and, critically, in articulating

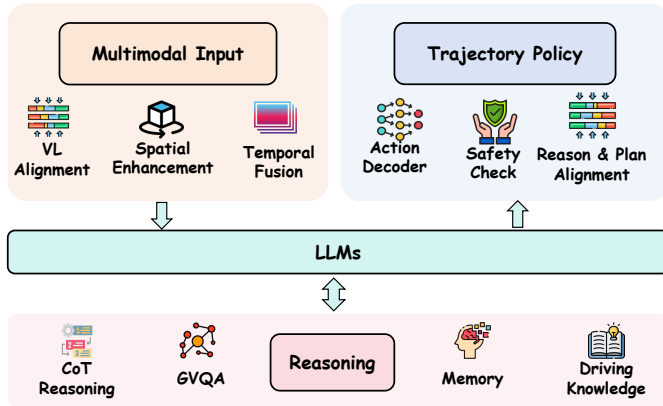


Fig. 5: Architecture of VLMs for E2E autonomous driving, including Multimodal Input (Sec. 3.1.1 and Sec. 3.1.2), Reasoning (Sec. 3.1.3) and Trajectory Policy (Sec. 3.1.4).

the rationale behind their decisions. This chapter presents a systematic survey of this nascent field, dissecting its *model architectures* (shown in Fig. 5) like spatiotemporal modeling and vision-language alignment; examining advanced *learning strategies* designed to transcend the limitations of behavioral cloning; and analyzing the *efficiency* optimization for resource-constrained deployment. Notably, this section is intentionally focused on VLM-native architectures that directly process sensory inputs, as hybrid frameworks that integrate VLMs within conventional E2E pipelines will be systematically treated in the subsequent chapter.

3.1 Model Structure

The architecture of a VLM is the principal determinant of its cognitive capabilities in autonomous driving. Contemporary systems largely converge on a canonical multi-stage architecture, where parallel vision and text encoders first extract unimodal features, after which a dedicated alignment module projects the visual embeddings into the semantic space of the LLM. These fused, multimodal representations are then processed by the LLM’s decoder, which functions as the central cognitive engine for high-level scene interpretation, causal reasoning, and decision-making. Beyond pure VLMs, recent work has focused on enabling models to output more detailed actions or planning waypoints, evolving them into Vision-Language-Action (VLA) models.

To provide a systematic deconstruction of this architectural paradigm, our subsequent analysis is organized around four fundamental pillars: the mechanisms for vision-language alignment (Sec. 3.1.1); the encoding and fusion of spatiotemporal information (Sec. 3.1.2); the explicit and implicit reasoning paradigms employed (Sec. 3.1.3); and the strategies for translating abstract decisions into executable planning and action policies (Sec. 3.1.4).

3.1.1 Vision-Language Alignment

Vision-language alignment aims to bridge the modality gap, transforming high-dimensional visual features into a tokenized representation residing within the LLM’s semantic embedding space. Current solutions primarily bifurcate into two dominant paradigms. The first, *Direct Projection*

typically employs a lightweight projector, such as a Multi-Layer Perceptron (MLP) [110], to execute a direct mapping of visual feature vectors into the target language embedding space [111, 112, 113, 114, 115, 116, 117, 118, 119]. In contrast, the *Query-based Compression* paradigm, epitomized by the Q-Former [120], employs a small set of learnable query vectors to distill and summarize the most salient information in visual information [74, 121, 122, 123, 124, 125, 126].

Orthogonal to the design of the alignment architecture itself, a significant research thrust has emerged to develop task-specific objectives that semantically ground LLMs in the specialized representations of autonomous driving, most notably BEV maps. These efforts move beyond generic alignment and focus on creating bespoke training signals tailored to the geometric and physical realities of the driving environment. One prominent strategy involves learning a shared embedding space through carefully designed proxy tasks. For instance, BEV-TSR [127] employs a cross-modal retrieval objective to align visual BEV features with textual descriptions, while GPVL [128] pioneers a dedicated 3D vision-language pre-training task to establish a direct correspondence between visual constructs and linguistic concepts within the BEV space. A complementary approach seeks to simplify the linguistic representation of spatial concepts. MPDrive [118], for example, enhances cross-modal consistency by circumventing complex textual coordinates in favor of concise visual markers for spatial referencing, thereby reducing ambiguity. More recently, this principle of task-driven alignment has been extended to encompass more heterogeneous data modalities beyond standard visual inputs. The work on Prompting Multi-Modal Tokens [129] demonstrates the end-to-end fusion of visual and LiDAR inputs into unified, learnable tokens. Driving with LLMs [130] addresses the critical challenge of aligning object-level vectorized data by introducing a novel "vector captioning" task. This objective explicitly trains the model to translate structured numerical information (*e.g.*, trajectories, bounding boxes) into a representation that an LLM can natively process, bridging the gap between symbolic reasoning and continuous physical state descriptions.

3.1.2 Spatiotemporal Understanding

Autonomous driving is inherently a four-dimensional problem, requiring a system to not only understand the static 3D spatial environment but also to capture its dynamic evolution over time. Consequently, a VLM’s ability to effectively model both spatial geometry and temporal dynamics is a prerequisite for all meaningful downstream driving tasks.

3D Spatial Information Enhancement: To rectify the geometric deficiencies of standard VLMs, research has diverged into several distinct strategies, which can be broadly categorized by their approach to incorporating 3D information. One major thrust is explicit 3D representation learning via architectural innovations. This involves re-engineering the VLM backbone to directly process spatial geometry. For instance, OmniDrive [82] integrates explicit 3D positional encodings, Atlas [131] employs a DETR-style 3D perceiver to directly tokenize geometric priors, and S4-Driver [83] introduces a learnable sparse volume to efficiently project multi-view 2D features into a unified 3D space. A second strategy is multi-modal 3D fusion, which grounds the

VLM’s understanding in veridical geometric data from external sensors. Methods like LiDAR-LLM [132], DriveMLM [74], LMDrive [75], and BEVDriver [122] augment the visual modality by integrating precise LiDAR point clouds. A third explicit method involves injecting object-centric priors to structure the spatial reasoning process. Reason2Drive [76] uses a prior tokenizer to refine relative spatial descriptions, MPDrive [118] abstracts complex coordinate prediction into a simpler semantic task via marker-based prompting, and DriveMonkey [133] leverages a pre-trained 3D object detector to enhance perception in cluttered scenes. In contrast to these explicit methods, a compelling alternative philosophy pursues implicit 3D awareness via data scaling. This paradigm, exemplified by Cube-LLM [134], posits that massive-scale, diverse training data can enable the model to develop an emergent understanding of 3D geometry without bespoke 3D architectural components.

Temporal Information Enhancement: Building upon spatial understanding, capturing the dynamics of a driving scene is equally critical. Sce2DriveX [116], for instance, pioneers the use of long-horizon video inputs combined with global BEV maps to learn from extended event sequences. LaVida Drive [114] emphasizes the importance of temporal context for improving reactivity in rapidly changing traffic situations. Furthermore, TrackingMeetsLMM [135] explicitly integrates object tracking information to enhance the model’s ability to reason about the behavior of individual agents over time. For more efficient aggregation of long-term information, ORION [136] introduces a query-based temporal module, QT-Former, which leverages a memory bank and history queries to effectively store and extract essential historical scene context.

3.1.3 Reasoning

Reasoning is the core capability that allows an autonomous agent to transition beyond simple, reactive behaviors, enabling it to make deliberate decisions. The emergence of LLMs and VLMs has catalyzed significant research into equipping autonomous driving systems with these sophisticated reasoning abilities. These approaches often draw inspiration from human cognition by integrating mechanisms for memory retrieval, structured logical inference, and grounding in external knowledge. This section systematically surveys these emerging approaches.

Reasoning with memory: This paradigm aims to emulate the human ability to leverage past experiences. Research in this area has largely bifurcated into two thrusts, distinguished by the temporal scale and strategic purpose of the memory. The first conceptualizes memory as a long-term repository to drive policy evolution and lifelong adaptation, enabling agents to generalize beyond their initial training. Works such as Drive Like a Human [137], DiLu* [138], LeapVAD [139], LeapAD [77], and Drive as You Speak [140] exemplify this by storing extensive driving episodes or user preferences, while Actor-Reasoner [141] builds an explicit experience database. In contrast, the second thrust utilizes memory as a short-term cognitive workspace to maintain task coherence and ground interactions in a dynamic context. This is evident in systems that track multi-step instructions (Agent-Driver [78]), manage long-horizon

navigation and dialogues (DriveVLM [142]), or maintain a consistent conversational history with the user [143].

Chain-of-Thought (CoT) Reasoning: CoT endows large models with the capacity for structured reasoning, decomposing complex tasks into intermediate, causally-linked steps. Within autonomous driving, this principle has manifested in several distinct research thrusts. The foundational approach, *Structured Logic CoT*, formulates robust driving strategies by decomposing the problem into a static, logically coherent sequence of sub-problems, as exemplified by works like Dolphins [144], CoT-Drive [145], ReasonPlan [146], X-Driver [147], and EMMA [80]. This static paradigm is naturally extended by *Spatiotemporal CoT*, which reasons over dynamically evolving visual information, a technique leveraged by FSDrive [148]. Diverging from these internal reasoning paradigms, *Tool-Augmented CoT* repurposes VLMs as semantic orchestrators that invoke specialized external modules, as demonstrated in Agent-Driver [78], AgentThink [149], Receive-Reason-React [150], and further advanced into an active perception framework by DriveAgent-R1 [151]. Subsequent research has focused on enhancing the reasoning process itself to address practical limitations. *Dynamic CoT*, for instance, tackles computational efficiency by enabling adaptive reasoning complexity, switching between "fast thinking" and "slow thinking" as in AutoVLA [84]. Meanwhile, to bolster reliability, methods like AutoDrive-R² [152] introduce an introspective validation loop, compelling the model to self-reflect on the plausibility of its own logic, thereby enhancing the fidelity of the final decision.

Graph Visual Question Answering (GVQA): To better emulate the structured, multi-step reasoning process of human drivers, some research leverages Graph Visual Question Answering (GVQA). Instead of relying on a monolithic reasoning process, GVQA-based methods first represent the traffic environment as a structured relational graph, then decompose the complex driving decision into a logically-ordered sequence of questions and answers over that graph, creating an explicit pathway for reasoning. For instance, DriveLM [79] explicitly mimics the human cognitive sequence of perception, prediction, and planning by constructing a chain of VQA pairs. This structure compels the model to first identify key actors and then reason about their interactions before forming a final driving decision. Similarly, SimpleLLM4AD [153] decomposes the driving task into a cascade of VQA pairs. By transforming the reasoning task into an explicit, step-by-step inquiry over a structured scene representation, this approach makes the model’s decision-making process more transparent and causally grounded.

Retrieval-Augmented Generation (RAG): To overcome the limitations of the model’s inherent parametric knowledge, RAG-based methods dynamically retrieve information from external knowledge bases at inference time to augment their decision-making. For example, RAG-Driver [112] achieves few-shot generalization by retrieving similar driving cases, while Driving with Regulation [154] ensures compliant behavior by retrieving relevant traffic rules.

Driving Knowledge Integration: In contrast to RAG’s dynamic retrieval of external knowledge at inference time, an alternative technical route seeks to permanently embed domain knowledge into the model during the training phase. These methods integrate expert knowledge, such

as traffic laws and physical principles, directly into the model’s architecture or training data. For instance, Hybrid Reasoning [155] fuses sensor data with an understanding of driving rules, and WiseAD [156] develops a knowledge-augmented VLM. These studies demonstrate that infusing models with explicit domain knowledge can significantly improve their reliability.

Reasoning for Interpretability: A key application for the reasoning ability is to enhance the interpretability of autonomous driving systems, which is a prerequisite for safety certification and public trust. The inherent ability of VLMs to articulate their reasoning in natural language offers a potential for enhancing model interpretability, thereby addressing the long-standing "black box" problem. VLAAD [123] trains the model’s explanatory abilities by building a specialized visual instruction tuning dataset. ADAPT [157] concentrates on generating explanations aligned with actions, outputting both the driving maneuver and its rationale. And Explanation for Trajectory Planning [121] aims to generate forward-looking justifications for future planned trajectories by projecting the planned trajectory as visual information onto the camera image and then having the VLM generate a natural language description of its intent.

3.1.4 Planning and Action Policy

Upon completion of the perception and reasoning phases by a VLM, the terminal challenge lies in transducing high-level semantic intentions into safe, precise, and executable actions. The current two primary technical paths either task the LLM with directly generating control outputs or use it as a high-level planner that guides planning and action module, thereby mitigating its poor numerical and spatial control capabilities. This section will focus on the latter "LLM + Action Module" paradigm, systematically reviewing its developments in *Action Head Design*, *Security Mechanisms*, and *Reasoning and Planning Alignment*.

Action Head Design: The action head serves as the critical interface for translating a VLM’s semantic decisions into the vehicle’s physical actions, fundamentally determining the execution quality of the driving policy. Two primary design paradigms currently exist. The first connects the VLM with a Controller, aiming to map high-level decisions to vehicle control commands in a tightly coupled manner. In practice, some works employ a "VLM + MLP + PID" combination, as seen in LMDrive [75], CarLLaVA [113], SimLingo [158], and DriveGPT4-V2 [85]. Others adopt a "VLM + MPC" paradigm, combining the VLM with Model Predictive Control (MPC), as explored in LanguageMPC [159] and VLM-MPC [160]. Empowering [161] takes this a step further by creating a feedback loop where the LLM provides safety constraints to the MPC, and the feasibility of the MPC solution is fed back to validate the LLM’s decision.

The second paradigm pairs the VLM with a Trajectory Planner, shifting the model’s output from direct control commands to a complete trajectory. In terms of implementation, some works employ rule-based planning. For example, DriveMLM [74] and ChatGPT as Your Vehicle CoPilot [143] use the VLM for high-level decision-making and then invoke a rule-based motion planning module. Other works explore generative planning by leveraging diffusion models to produce multiple candidate trajectories (e.g., DriveMoE [86],

Diff-VLA [162], and ReCogDrive [163]), thereby enhancing planning flexibility and robustness.

Security Check Mechanisms: Due to the inherent uncertainty of LLMs, often referred to as the "hallucination" problem, establishing reliable security check mechanisms before action execution is crucial. Current methodologies primarily diverge into two paradigms. The first, *post-hoc verification and refinement*, entails validating or optimizing a generated decision prior to its execution, encompassing methods like CALMM-Drive [164], which filters multiple LLM-proposed candidates through a hierarchical refinement process; PlanAgent [165], which employs simulation-based evaluation for waypoint safety; and LeapAD [77] with LeapVAD [139], which leverage memory-based reflection to iteratively refine actions against historical performance. The second paradigm, *in-training feedback and guidance*, shifts the focus from extrinsic correction to intrinsic safety alignment by incorporating feedback signals directly into the training phase. FeD [166] exemplifies this approach, utilizing an LLM to provide fine-grained linguistic feedback that guides the agent’s learning process.

Reasoning and Planning Alignment: Ensuring that high-level reasoning ("why") is consistent with low-level planning ("how") is key to improving system trustworthiness. A common pain point is the mismatch between the generated CoT explanation and the final driving action. To address this, SimLingo [158] incorporates a language-action alignment mechanism. RDA-Driver [119] enforces consistency by introducing a Reasoning-Decision Alignment Constraint during training. To fundamentally bridge the gap between the semantic reasoning space and the numerical planning space, ORION [136] introduces a Generative Planner. This module is designed to learn a unified latent representation, enabling the VLM’s reasoning output to directly and effectively instruct the generation of precise trajectories, thereby achieving a deep alignment and joint optimization of the two spaces.

3.2 Learning Strategies

The realization of robust and generalizable autonomous driving systems depends not only on advanced model architectures but also on the adoption of more effective learning strategies for navigating complex driving scenarios. To address the inherent limitations of imitation learning (discussed in Sec. 2.3), this section discusses three primary learning strategies employed within the VLM for Autonomous Driving paradigm, including Knowledge Distillation (Sec. 3.2.1), Reinforcement Learning (Sec. 3.2.2) and Data-Efficient Learning (Sec. 3.2.3).

3.2.1 Knowledge distillation

Similar to the knowledge distillation approach adopted by conventional E2E autonomous driving discussed in Sec. 2.3.1, the VLM-centric E2E paradigm also employs privileged agents as teacher models to provide dense supervision signals. For instance, by distilling knowledge from a privileged MLLM and using feedback-guided fine-tuning, FeD [166] introduces a feedback-driven framework that allows a model to analyze the causes of its action failures and correct planned trajectory, which results in high-performance open-

and closed-loop capabilities. Additionally, knowledge distillation is utilized to mitigate catastrophic forgetting issues of VLMs for AD. VLM-assisted [167] combines a VLM with selective memory replay and knowledge distillation. This allows a previously trained model to serve as a teacher, guiding subsequent tasks and thereby minimizing catastrophic forgetting during the continual learning process for autonomous driving VQA tasks.

3.2.2 Reinforcement Learning

As previously discussed in Sec. 2.3.2, reinforcement learning (RL) is increasingly being leveraged to push conventional E2E autonomous driving beyond the limitations of imitation learning for substantial performance gains. This is equally common in the VLM-centric E2E paradigms. AlphaDrive [168] and Poutine [169] adopt a two-stage training strategy: pre-training on driving-related QA tasks followed by reinforcement fine-tuning with Group Relative Policy Optimization. Guided by rewards that combine performance scores and format constraints, the RL stage improves overall model capability. ReCogDrive [163] integrates diffusion planners with VLAs and fine-tunes them with RL, achieving more effective planning. AutoVLA [84] further combines RL with Chain-of-Thought (CoT) reasoning, where pre-training on mixed CoT datasets and reward-based constraints on CoT length enable dynamic reasoning.

3.2.3 Data-Efficient Learning

The advancement of Vision-Language-Action models (VLAs) for autonomous driving is often constrained by the resource-intensive process of high-quality data annotation. To mitigate this dependency on costly annotated data, researchers are exploring various data-efficient learning strategies, such as self-supervised and semi-supervised paradigms, which enable models to leverage abundant unlabeled data alongside limited annotated samples.

For instance, LDM [115] generates pseudo-answers for unlabeled data using a model trained with limited annotated data, and obtains more reliable VQA data through self-consistency refinement, finally achieving comparable performance with only 5% of the labeled data. ReasonPlan [146] enhances a VLA model’s spatial understanding by jointly training on a self-supervised Next Scene Prediction (NSP) task and a supervised Decision Chain-of-Thought (DeCoT) process. This method aligns visual representations with actionable contexts, yielding more interpretable and causally grounded decisions. Moreover, considering that most VLA training currently does not utilize 3D spatial annotated data, S4-Driver [83] proposes a scalable, self-supervised motion planning algorithm that seamlessly transforms 2D visual representations from MLLMs into 3D space using an innovative sparse volume strategy, achieving superior performance and demonstrating excellent scalability.

3.3 Efficiency

While VLM-based systems demonstrate immense cognitive potential, their high computational overhead remains a critical bottleneck for practical application. Addressing this efficiency challenge has spurred research along several distinct technical philosophies, as detailed below.

3.3.1 Model Distillation for Efficiency

To address the performance-efficiency trade-off, model distillation has emerged as a key technique. Recent works such as CoT-Drive [145] and DSDrive [125] have made significant explorations in this direction. CoT-Drive [145] utilizes CoT prompting to distill the structured scene understanding process from a large online LLM into a lightweight edge model, thereby preserving core reasoning abilities while reducing computational overhead. Building upon this, DSDrive [125] further focuses on the alignment of reasoning and planning. It introduces a coordination module that aligns the model’s reasoning and actions by using the final waypoints as a supervisory signal for the reasoning process.

3.3.2 Model Structure Optimization for Efficiency

Directly modifying the model structure of VLMs is another key pathway to enhancing their operational efficiency. One class of methods focuses on dynamically adjusting computational resource allocation. For instance, AD-EE [170] introduces an early-exit framework that allows the model to terminate inference prematurely once a confident prediction is reached. DynRsl-VLM [171], on the other hand, employs a dynamic resolution module to maintain high resolution for critical regions of interest while reducing it for background areas, thereby minimizing visual processing computation while preserving essential details. DriveMoE [86] employs Mixture-of-Experts (MoE) to tackle challenges of multi-view information redundancy. A second category of structural optimization focuses on reconfiguring the model’s inference workflow. AsyncDriver [172], for instance, proposes an asynchronous reasoning framework that decouples the high-frequency real-time planner from the low-frequency LLM reasoning process, significantly reducing the computational burden imposed by the LLM.

3.3.3 Data-centric Approaches for Efficiency

Beyond model optimization, optimizing data representation is another key technical route. Some works focus on structuring the model’s decision outputs. For instance, ReCogDrive [163], Senna [81], and VERDI [173] convert free-form textual decisions into specific formats. In a more comprehensive approach, FastDrive [174] achieves over a 10x inference speedup on a compact VLM without model compression by introducing a structured dataset (NuScenes-S) that eliminates redundant information from the input. Other methods employ token pruning strategies, such as the complexity-adaptive pruning approach used in AutoPrune [175], which significantly reduces computational overhead to accommodate the computational constraints of real-world autonomous driving scenarios.

4 VLM EMPOWERS CONVENTIONAL END-TO-END AUTONOMOUS DRIVING

As established in the previous chapter, conventional end-to-end models and Vision-Language Models present a compelling complementarity: the former acts as a fast and precise reactive body but suffers from a cognitive gap, while the latter serves as a deliberative cognitive brain but is hindered by an action gap. To harness the strengths of both,

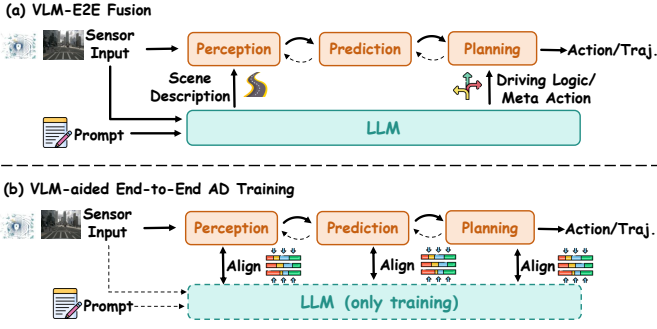


Fig. 6: Main innovations of the fusion of VLMs and E2E autonomous driving, including VLM-E2E Fusion (Sec. 4.1), VLM-Aided E2E Training (Sec. 4.2).

researchers in academia and industry are actively exploring hybrid architectures that inject the VLM’s reasoning capabilities into the conventional E2E pipeline, aiming to create an agent that is both agile and profound. This chapter analyzes this emerging field by categorizing its mainstream into *online coordination via hierarchical fusion*, and *offline knowledge transfer via VLM-aided training*, as shown in Fig. 6.

4.1 Online coordination via Hierarchical Fusion

The predominant paradigm for integrating VLMs with conventional end-to-end systems is through a hierarchical fusion architecture, as illustrated in Fig. 1 (d). This design conceptually decomposes the driving task, positioning the VLM as a high-level cognitive engine for strategic decision-making, while the conventional E2E model acts as a low-level execution unit for physical trajectory generation. In this online coordination framework, a fusion interface connects the VLM’s cognitive guidance with the E2E’s physical execution, typically operating at the perception or planning level. However, this design also introduces core challenges such as inference latency and error propagation, which remain a key research focus in the field.

4.1.1 Fusion at the Perception Level

A foundational yet critical strategy is to perform fusion at the perception level, with the core objective of leveraging the VLM’s semantic understanding to perform semantic augmentation of the E2E model’s perceptual representations. VLM-E2E [176] conducted pioneering exploration in this direction, using a VLM to generate text prompts highly relevant to driver attention, which then serve as semantic supervision signals for deep fusion with BEV features. While VLM-E2E [176] primarily addresses the challenge of what to fuse, a parallel challenge lies in the mechanism of how to fuse, specifically, how to efficiently integrate a powerful but slow VLM with a high-frequency, responsive E2E. To tackle this problem of asynchronous integration, NetRoller [177] proposes a sophisticated adapter framework. It does not wait for the VLM to complete its full inference. Instead, it employs a novel “instantaneous first-token, all-layer acquisition” mechanism to efficiently capture the instruction-rich information embedded in the intermediate-layer variables of the VLM, allowing the E2E system’s perception capabilities to benefit from the VLM’s deep reasoning in real-time while circumventing its inherent high latency.

4.1.2 Fusion at the Planning Level

The most prevalent and direct implementation of the hierarchical fusion architecture occurs at the planning level. The core of this approach is to use the VLM’s high-level decisions to guide or constrain the E2E model’s low-level trajectory generation, aiming to enhance the semantic rationality and foresight of the planning.

The paradigmatic prototype for this technical direction was established by early works like DriveVLM [88] and further formalized by methods such as Senna [81]. For instance, Senna [81] establishes a decomposed architecture by using a VLM to generate high-level commands, which then serve as a “meta-action” to condition a downstream trajectory planner. Building on this, some later works focused on emulating human cognition, such as DME-Driver [178] that incorporates human driving experience, and FASIONAD [89] which builds a dynamic switching system inspired by “Thinking, Fast and Slow” theory. Others centered on architectural optimization, with VLAD [179] validating the effectiveness of fusion through simple system integration and LeAD [180] designing a dual-rate architecture for complex scenarios. Collectively, these explorations have advanced the VLM-guided planning paradigm from early unidirectional command passing to a more adaptive and dynamic, context-aware cognition-action loop.

4.1.3 Synergistic Fusion across Perception and Planning

Building on single-point fusion, subsequent research has begun to explore deep synergistic models, primarily through three pathways. The first is bidirectional information exchange, exemplified by DriveVLM [88], where the E2E system’s perception outputs are cross-validated by the VLM, which in turn provides a reasoned trajectory hint back to the planner. A second path is deep architectural sharing, as seen in SOLVE [90], which employs a shared vision encoder for both the VLM and E2E model and uses VLM predictions to initialize the real-time planner. The third path extends the synergistic concept to enhance interpretability. Hint-AD [181], for instance, uses a post-hoc VLM to receive intermediate outputs from the E2E pipeline and generate aligned natural language explanations. These explorations signify that VLM-E2E architectures are evolving from simple module concatenation to deeply integrated organic symbiosis, aimed at building an intelligent system with more efficient information flow and tighter functional coupling.

4.2 Offline Knowledge Transfer via VLM-aided Training

A distinct and highly promising technical direction, starkly different from online coordination, is to leverage the capabilities of VLMs exclusively during the training phase, as illustrated in Fig. 1 (e). In this paradigm, termed VLM-Aided Training, the VLM acts as a powerful teacher, providing rich, multi-faceted supervision signals. Through knowledge distillation, the cognitive abilities of the VLM are internalized into the parameters of the conventional E2E model. The core advantage of this method is its ability to significantly enhance the E2E model’s performance and generalization by internalizing the VLM’s cognition, all without incurring any additional inference cost or deployment complexity. However, while this paradigm is efficient, its performance

ceiling is constrained by the knowledge coverage of the teacher VLM and the learning capacity of the student model. Ensuring the unbiasedness of the distilled knowledge and validating the generalization of the offline-learned policy in the open world remain key challenges for this paradigm.

4.2.1 Planning and Action Alignment

A primary application of this paradigm is to align an E2E model’s latent representations and decision outputs with high-level planning intentions derived from a VLM, ensuring the generated trajectories are grounded in a semantically meaningful understanding of the required maneuver. For instance, VLM-AD [182] utilizes a VLM teacher to generate structured action labels (*e.g.*, "turn", "lane change") as supplementary supervision, which forces the E2E model’s internal features to align with these explicit commands. Similarly, DIMA [183] aligns the E2E model’s structured latent representations with the understanding of a teacher model, thereby pre-infusing the planner’s input with high-level reasoning. In these approaches, the teacher VLM is discarded after training, ultimately yielding an enhanced conventional E2E model with more semantically consistent decision-making at zero additional inference cost.

4.2.2 Perception, Planning, and Reasoning Alignment

Building upon the alignment of planning and action, a more comprehensive approach extends the scope of alignment to the entire decision-making chain, striving to achieve a holistic alignment between the VLM’s structured reasoning and the entire perception-prediction-planning pipeline of the E2E model. The goal of this full-stack alignment is to ensure that the model’s final decision originates from a causally coherent and logically sound process, from initial perception to final action. For example, VERDI [173] employs a modular alignment approach, using VLM-generated CoT outputs to separately supervise the intermediate features of each sub-module in the perception-prediction-planning pipeline. In contrast to modular supervision, VLP [184] focuses on the fine-grained alignment of key representations, such as aligning the E2E model’s BEV features with an idealized BEV representation from an LLM. To construct a more systematic end-to-end constraint, ALN-P3 [185] proposes a unified co-distillation framework that defines explicit alignment mechanisms covering the perception, prediction, and planning stages, mandating that visual representations throughout the E2E stack must be consistent with the corresponding linguistic outputs from the VLM. Although their technical paths vary, these methods share a common goal: to fundamentally enhance the robustness and trustworthiness of the E2E model at zero inference cost, ensuring it not only performs the correct action but does so for the right reasons.

5 DATASET AND BENCHMARK

This section details the datasets and benchmarks commonly used for evaluating general end-to-end autonomous driving systems. Besides, we also provide a detailed comparison of SOTA methods on popular benchmarks such as nuScenes [105] and CARLA [186].

5.1 Dataset without Language Annotation

Conventional end-to-end autonomous driving adopts a sequential pipeline comprising perception, prediction, and planning. Since this paper focuses on analyzing planning-oriented end-to-end autonomous driving, this section primarily introduces datasets for evaluating the planning performance of models under open- and closed-loop protocols.

For open-loop evaluation, the most commonly used dataset is nuScenes [105]. This evaluation process is operated on logged sensor data with frozen scene evolution under standardized offline metrics. Though scaling well and highly reproducible, it breaks the feedback between the planner and the environment, which can overestimate performance. Closed-loop simulation addresses this limitation by restoring interaction and controlling both the information exposed to the planner and the dynamics of the environment, yielding a realistic and repeatable assessment. Within this regime, CARLA [186] provides configurable weather, illumination and traffic, together with metrics that combine route completion, safety, and infraction penalties for comprehensive evaluation. Bench2Drive [218] extends CARLA with task-complete scenarios, enabling more reliable and fine-grained closed-loop evaluation.

Closed-loop evaluation requires high computational overhead, while open-loop evaluation cannot reflect real-world closed-loop performance. Therefore, NAVSIM [54] utilizes a non-interactive simulator and simulation-based metrics to demonstrate real-world performance. Essentially, it uses an open-loop approach to perform nearly closed-loop evaluation, balancing reliability and efficiency, and is gradually becoming a more mainstream evaluation method.

5.2 Dataset with Language Annotation

In recent years, numerous vision-language (VL) datasets for autonomous driving have been proposed to improve model performance, provide natural language instructions to vehicles, and enhance interpretability. These motivations have gradually evolved into various specific VL tasks. Early datasets relied entirely on manual annotation, which made data release costly and yielded sparse or coarse descriptions. For example, several works built upon BDD100K [219], such as BDD-X [187] and BDD-OIA [190] for explaining driving behavior, as well as HAD [188] and DRAMA [25] based on large collections of real-world videos.

Early studies often struggled due to the limited quantity and quality of datasets. However, since the advent of LLMs/VLMs, the cost of annotation has been greatly reduced, and the descriptions of scenes and objects have been significantly enriched. As shown in Tab. 2, vision-language datasets have witnessed a rapid surge in quantity, scale and diversity. Meanwhile, with the release of nuScenes[105] as a high-quality dataset, many studies focus on constructing high-quality vision-language datasets based on it. Therefore, as illustrated in the Fig. 2, we divide the subsequent works into four categories according to their sources. And these works evaluated language outputs with BLEU [220], CIDEr [221], GPT-4-based scores [222] and other NLP metrics [223, 224].

Datasets Fully Derived from nuScenes: Talk2Car [189] first provides object-referral annotations in nuScenes using

TABLE 2: Comparison of Vision-Language Autonomous Driving Datasets.

- **Dataset Type:**  real world data,  simulated data.
- **Image Views:**  denotes mixed perspectives
- **Task:**  Traffic Scene Understanding,  Meta Action,  Ego Trajectory Planning,  Future Prediction,  Visual Grounding,  Tracking.
- **Labeling Tool:**  LLM,  VLM,  Human Annotated,  Rule-based,  CM using Corruption Models.

Dataset	Year	VL	Type	Videos	Clips	Frames	Image Views	3D Tasks	VL Tasks	QA Pairs	Sources	Labeling Tool
BDD-X [187]	2018	✓		7K	–	–	1	–	 	26K	BDD	
HAD [188]	2019	✓		5.6K	–	–	1	–	 	45K	HDD	
Talk2Car [189]	2019	✓		–	–	9K	6			12K	nuScenes	
BDD-OIA [190]	2020	✓		23K	–	–	1	–	 	23K	BDD	
SUTD-TrafficQA [191]	2021	✓		10K	–	–	1	–		62K	self-collected	
DiveMLM [74]	2023	✓		–	–	–	4	–	  	–	Carla	
Reason2Drive [76]	2023	✓		–	–	–		–	  	600K	nuScenes, Waymo, ONCE	
Refer-KITTI [192]	2023	✓		18	–	–	1	–		818	KITTI	
DRAMA [25]	2023	✓		18K	–	–	1	–		18K	self-collected	
DriveLM-nuScenes [79]	2024	✓		–	–	4.8K	6	–	   	443K	nuScenes	
DriveLM-CARLA [79]	2024	✓		–	–	183K	4	–	   	3.7M	Carla	
LMDrive [75]	2024	✓		–	–	3M	4	–	 	528K	Carla	
DriveCoT [193]	2024	✓		–	–	36K	6	–	  	36K	Carla	 
NuScenes-QA [194]	2024	✓		–	–	34K	6	–		460K	nuScenes	
NuScenes-MQA [195]	2024	✓		–	–	34K	6	–		1.4M	nuScenes	 
Talk2BEV [196]	2024	✓		–	–	1K	6	–		20K	nuScenes	 
DriveGPT4 [72]	2024	✓		7K	–	–	1	–		56K	BDD	
ContextVLM [197]	2024	✓		–	–	1.5K		–		35K	nuScenes, KITTI, self-collected	 
LingoQA [126]	2024	✓		–	28K	–	1	–	 	419K	self-collected	 
Rank2Tell [198]	2024	✓		116	–	–	3		 	5.8K	self-collected	
MAPLM-QA [199]	2024	✓		–	–	14K	7	–		61K	self-collected	
NuInstruct [200]	2024	✓		–	–	12K	6	–	  	91K	nuScenes	 
DriveVLM [88]	2024	✓		1K	–	–	–	–	  	–	self-collected	
SURDS [201]	2024	✓		–	–	33K	6	–		50K	nuScenes	
nuScenesR ² -6K [152]	2025	✓		–	–	6K	1	–	  	6K	nuScenes	 
Bench2ADVLM [202]	2025	✓		–	–	2M	6	–	   	–	Bench2Drive	 
VLADBench [203]	2025	✓		–	3K	2K		–	  	12K	GTSDb, JAAD, PIE, HAD, nuScenes, SODA, ONCE, Argoverse2, CODA, DRAMA, RS10K, LingoQA	
NuInteract [204]	2025	✓		–	–	34K	6		  	1.5M	nuScenes	 
Drive-R1 [205]	2025	✓		–	–	4K	6	–	 	4K	nuScenes	 
DriveAction [206]	2025	✓		–	2.6K	–	1	–	 	16K	self-collected	
STSBench [207]	2025	✓		–	1.2K	–	6	–		10K	nuScenes	 
DRAMA-ROLISP [208]	2025	✓		18K	–	–	1	–		18K	DRAMA	
WOMD-Planning-ADE [83]	2025	✓		103K	–	–	8	–	 	400K	WOMD	
ImpromptuVLA [209]	2025	✓		–	–	80K		–	   	560K	nuScenes, argoverse, idd, kitti, mappillary, navsim, once, waymo	
DriveBench [210]	2025	✓		–	–	19K		–	 	20K	DriveLM	
SimLingo [158]	2025	✓		–	–	1M	–	–	  	28M	Carla	
WOMD-Reasoning [211]	2025	✓		–	–	63K	8	–	  	3M	WOMD	
OmniDrive [82]	2025	✓		–	–	34K	6	–	  	544K	nuScenes	 
CODA-LM [212]	2025	✓		–	–	10K	1	–	 	21K	CODA	
CoVLA [213]	2025	✓		10K	–	–	1	–	  	6M	self-collected	 
nuPrompt [214]	2025	✓		–	–	34K	6			40K	nuScenes	 
Robusto-1 [215]	2025	✓		200	–	–	1	–		3K	self-collected	 
DrivingVQA [216]	2025	✓		–	–	4K	1	–		4K	self-collected	 
DriveLM-o1 [217]	2025	✓		–	–	2K	6	–	 	5K	nuScenes	

natural language commands. DriveLM [79] introduces a Graph VQA task with template-based labeling, while subsequent works [194] [195] [204] [196] further expand QA pairs. DriveBench [210] novelly evaluates whether VLMs truly rely on visual cues. OmniDrive [82], NuPrompt [214], SURDS [201], STSBench [207] and NuInstruct [200] exploit nuScenes' 3D spatial and temporal information for more comprehensive QA and instruction-following. In addition, several works annotate reasoning chains for ego-vehicle behaviors [152, 205, 217], with Drive-R1 [205] and nuScenesR²-6 [152] also providing trajectory point annotations.

Datasets Constructed from Other Open-Source Data:

Beyond nuScenes, many datasets leverage other high-quality autonomous driving corpora. Refer-KITTI [192] defines a referring understanding task on KITTI [225]. DriveGPT4 [72] builds a visual instruction tuning dataset on BDD [219] with ChatGPT-generated annotations, while DRAMA-ROLISP [208] enriches visual information in DRAMA [25]. On WOMD [226], WOMD-Planning-ADE-Benchmark [83] focuses on spatio-temporal visual representations, and WOMD-Reasoning [211] targets traffic rule-induced interactions. Like ImpromptuVLA [209], these

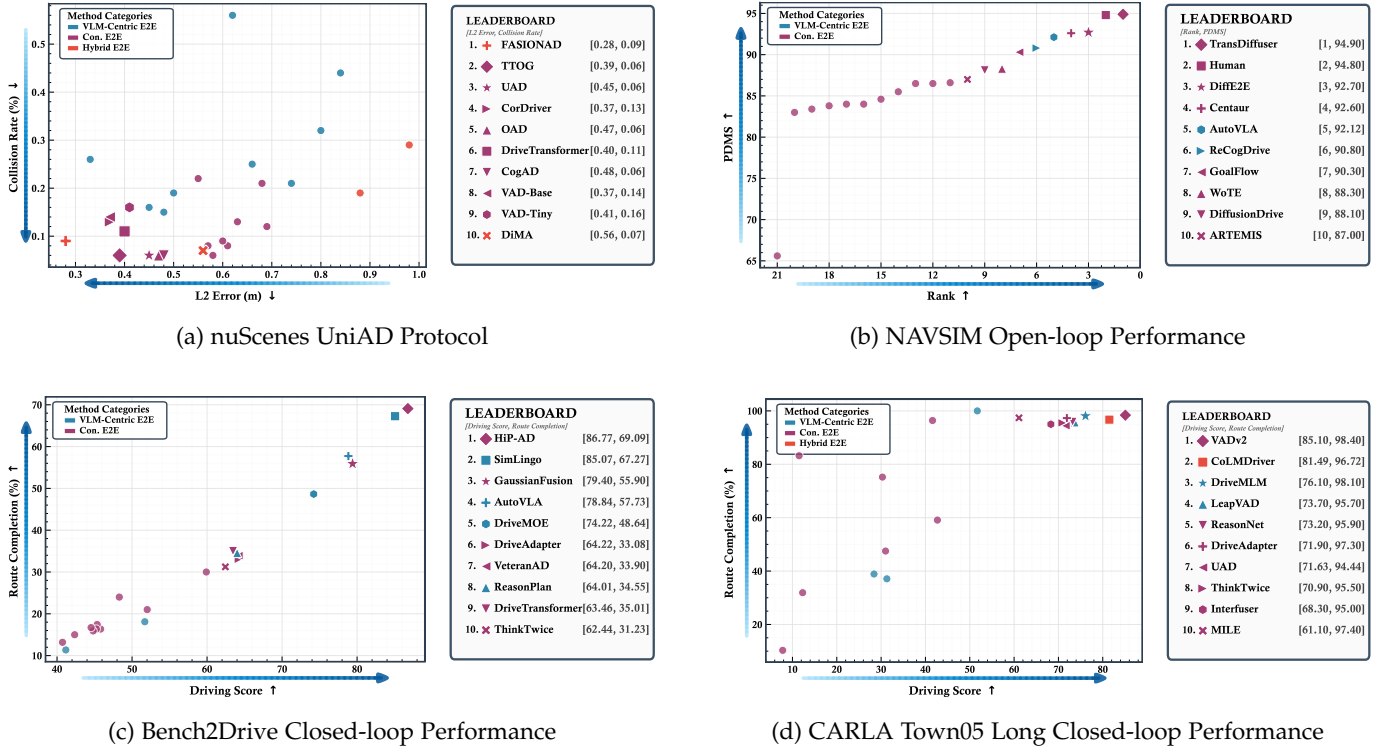


Fig. 7: Open-loop and Closed-loop Performance Comparison in nuScenes [105], NAVSIM [54], Bench2Drive [218] and CARLA [186]. For each dataset, we use different colors to represent three E2E paradigms (Conventional E2E, VLM-centric E2E and Hybrid E2E). The top ten performing methods are listed on the right side of the graph, each with a unique legend entry. All other methods are represented as small dots. The ranking criteria vary across datasets: for the nuScenes leaderboard, the ranking score is calculated by summing the min-max normalized collision rate and L2 error. In contrast, rankings for NAVSIM, Bench2drive and CARLA are based directly on their comprehensive driving scores. Finally, we labeled the direction representing better performance along the coordinate axes.

works [76] [197] [203] all draw upon multiple datasets to construct a comprehensive dataset.

Datasets Collected Independently: Some works collect their own data instead of relying solely on existing benchmarks. LingoQA [126], MAPLM [199], and CoVLA [213] gather large-scale driving images to broaden AD coverage. Robusto-1 [215] and Rank2Tell [198] independently acquire hundreds of driving videos. CODA-LM [212] and DriveVLM [88] explicitly search for corner cases to alleviate class imbalance in open datasets. In parallel, DriveAction [206] leverages real users’ driving data, and DrivingVQA [216] uses items from driving theory examinations to build a VQA dataset.

Datasets Generated from CARLA Simulation: Another type work constructs simulated closed-loop data in CARLA [186]. DriveMLM [74] first annotates such data for VLM training. DriveLM [79] and LMDrive [75] later release their simulated data and language annotations to enhance task diversity. DriveCoT [193] exploits sensor streams and control decisions in CARLA to annotate chain-of-thought labels, explicitly exposing the reasoning process. Bench2ADVLM [202] builds on Bench2Drive [218] routes to generate 220 threat scenarios with a self-reflective scenario generation module, enabling customized evaluation, while SimLingo [158] is specifically designed to train models for robust language-action alignment.

5.3 Performance Analysis

This section analyzes the open-loop and closed-loop performance of end-to-end autonomous driving models across several prominent benchmark datasets, including nuScenes [105], Bench2Drive [218], and CARLA Town05 Long [186], offering a comparative study of three different paradigms.

Open-loop Performance: Open-loop performance is primarily evaluated on the nuScenes [105] and NAVSIM [54] benchmark. The nuScenes employs two evaluation protocols: protocols adopted by ST-P3 [11] and UniAD [9], which both present two metrics: average L2 error within 3 seconds and average collision rate. We primarily compile statistics from UniAD’s evaluation protocols to ensure a fair comparison. NAVSIM’s statistics are mainly based on its designed Predictive Driver Model Score (PDMS), a comprehensive metric that considers safety, comfort, and efficiency.

An examination of the leaderboard in Fig. 7a reveals that the best method belongs to the hybrid E2E paradigm. This suggests that the integration of VLMs provides conventional E2E models with valuable world knowledge, thereby enhancing their performance, particularly in handling corner cases. Moreover, most of the top ten methods on nuScenes employ conventional end-to-end algorithms, demonstrating the superiority of conventional end-to-end approaches in numerical trajectory prediction. Surprisingly, the best methods on the NAVSIM leaderboard in Fig. 7b

outperform human driver evaluations, demonstrating the advanced capabilities of current autonomous driving systems. Additionally, we discover two VLA-based approaches achieving strong results on NAVSIM, while other VLA methods on nuScenes are gradually catching up to or even surpassing conventional end-to-end approaches. This trend demonstrates the significant potential of the VLA approach for realizing more robust planning capabilities.

Closed-loop Performance: For closed-loop performance analysis, we choose the Bench2Drive [218] benchmark and challenging Town05 long in CARLA [186], as illustrated in Fig. 7c, 7d. The two main metrics are Route Completion and Driving Score. Route Completion indicates the percentage of the route distance completed by an agent, and Infraction Score indicates the degree of infractions happening along the route. Driving Score serves as the product between the Route Completion and the Infraction Score, which is the main metric for ranking.

While SOTA methods demonstrate commendable robustness, evidenced by high performance in traffic rule adherence and closed-loop driving, the top route completion rate in Bench2Drive fails to exceed 70%, indicating that handling diverse and long-horizon routes remains a critical bottleneck. In the closed-loop leaderboard, the conventional end-to-end methods account for a larger proportion of the leading methods, especially in the Town05 long benchmark, this phenomenon is more obvious. This indicates that the VLA method still has considerable room for improvement in terms of closed-loop performance. That is, VLA needs to be able to clearly understand the impact of trajectory prediction on the environment, achieve more fine-grained control over trajectories, and reduce overfitting in trajectory prediction.

6 CHALLENGES

From a unified perspective on general end-to-end autonomous driving, this section will analyze the persistent challenges and critical problems that are expected to be addressed in future work.

6.1 Long-tailed Data Distribution

The data distribution of real-world driving scenarios is characterized by a typical long-tail problem, wherein a small subset of common scenarios constitutes the vast majority of the total data volume. This inherent imbalance limits the model’s generalization, leading to suboptimal or poor performance in complex or rare hazardous scenarios (*i.e.*, corner cases). Many researchers have attempted to solve this problem, yet most of the solutions have their own flaws.

Several approaches have been explored to mitigate the long-tail problem in autonomous driving data. A common strategy is to employ simulation or generative AI to create datasets of rare scenarios [227, 228, 229, 230]. The primary limitation of such methods is the "sim-to-real" gap, alongside challenges in ensuring the physical and causal fidelity of the generated content.

An alternative direction shifts focus from data generation to leveraging the intrinsic knowledge of large pre-trained models. VLMs, with their extensive world knowledge, present a compelling alternative. However, their application is often plagued by catastrophic forgetting, where

fine-tuning on specialized driving tasks leads to a degradation of their generalist knowledge, paradoxically hindering generalization [167]. While Reinforcement Learning (RL) offers an exploratory mechanism to discover novel behaviors beyond the training data, it remains hampered by significant challenges in sample efficiency, simulation fidelity, and reward design. More integrated frameworks like lifelong learning [231] aim to bridge this gap by continuously incorporating new driving data. A key bottleneck, however, is their dependency on a VLM to label these new scenarios, creating a recursive vulnerability.

6.2 Explainability

Explainability is the foundation for building deep trust between autonomous driving systems and passengers, and it is crucial for diagnosing potential model failures in the event of incorrect decisions.

Traditional end-to-end autonomous driving models primarily use abstract visual features as intermediate representations, which prevents them from outputting human-understandable language, leading to a lack of transparency and explainability in their decision-making process. To enhance the explainability of end-to-end models, some methods [91, 232, 233, 234] attempt to visualize the attention maps, representing the regions the model focuses on during decision-making. Other approaches [9, 70] utilize latent features for auxiliary visualization tasks, such as object detection, to improve interpretability. However, these methods still fall short of revealing the specific decision-making process and are susceptible to issues of spurious correlation. To further increase transparency, some techniques [157, 181, 235] feed the model’s intermediate representations into a language decoder to predict the rationale or reasoning process behind a decision. This approach, however, is highly dependent on the format of the intermediate outputs, limiting its adaptability, and its explanatory performance in complex scenarios remains constrained.

In contrast to traditional end-to-end autonomous driving, the VLA paradigm inherently possesses stronger explainability. It leverages Chain-of-Thought (CoT) reasoning to enhance its decision-making process, providing interpretable justifications for its driving decisions while performing precise motion planning. However, a challenge arises with Vision-Language Models (VLMs) where inconsistencies can occur between motion planning and language reasoning, which undermines their explainability in practical applications. RDA-Driver[119] addresses this by constructing positive and negative samples of matched and mismatched reasoning-decision pairs and employing a contrastive learning loss function for training. This forces the model’s internal assessment of the "reasoning process" to align with the actual quality of the "final decision." Nevertheless, this method requires additional dataset construction and does not explicitly teach the model the intrinsic causal relationship between reasoning and planning, indicating that there is still room for improvement.

6.3 Safety Guarantee

In real-world road scenarios, safety constitutes the paramount requirement for autonomous driving. Contemporary end-to-end autonomous driving systems primarily

improve safety through either (a) incorporating safe driving constraints during the training phase [29] or (b) implementing post-processing safety modules during inference to verify the safety of model-generated actions [9, 50]. However, constrained by the inherent complexity of real traffic environments and the intrinsic uncertainty of deep learning models, relying solely on safety constraints during training proves insufficient to ensure reliable safety guarantees. In contrast, supplementary safety verification modules can establish performance baselines to some extent, effectively suppressing high-risk behaviors caused by overly aggressive decision-making. Nevertheless, such modules simultaneously compromise the structural simplicity of end-to-end architectures and may induce excessively conservative strategies, thereby sacrificing opportunities for improved driving efficiency and user experience. Consequently, establishing a dynamic and quantifiable equilibrium between safety assurance and decision-making efficiency remains a critical challenge requiring in-depth investigation.

6.4 Real-time Efficiency

In autonomous driving, the capacity for real-time, rapid decision-making is critical for ensuring safety. However, the substantial inference latency of VLMs, stemming from their vast parameter counts and autoregressive generation processes, directly conflicts with this operational demand. To bridge this gap, various strategies have been developed to enhance computational efficiency as illustrated in Sec. 3.3, including model distillation, structural optimizations such as early exiting and mixture-of-experts, and data-centric methods designed to simplify data complexity. Nevertheless, these techniques can impose limitations on model robustness and reliability. For instance, the performance of a distilled student model is inherently capped by its teacher model, early exiting strategies carry the risk of erroneous judgments, and the reliance on structured data representations can diminish a model’s capacity for free reasoning. Consequently, achieving low-latency inference without compromising the driving robustness remains a critical and open problem.

7 FUTURE TRENDS

Considering the challenges and opportunities discussed, we analyze some crucial directions for future research that may have a broader impact in this field.

Reinforcement Learning. The paradigm of end-to-end autonomous driving is gradually shifting from the dominance of imitation learning to a hybrid approach that combines offline pre-training with online policy fine-tuning. In this trend, imitation learning continues to serve as an efficient pre-training method for rapidly acquiring deployable initial policies. At the same time, reinforcement learning (RL) extends beyond imitation by leveraging active exploration to gather experience outside expert demonstrations. This capability allows RL to address limited data coverage in long-tail and highly interactive scenarios. Moreover, through reward modeling, RL explicitly encodes objectives such as safety, efficiency, and compliance, thereby guiding policies toward long-term optimization [168, 169]. Together,

these advances provide strong evidence for the shift from passive imitation to goal-oriented decision optimization.

Foundation Models. A promising research direction is the large-scale pre-training of foundational models for autonomous driving, followed by rapid fine-tuning for specific scenarios. These foundation models [80, 87], often leveraging VLA architectures, utilize world knowledge and common-sense reasoning ability learned from massive, general-purpose datasets to empower driving tasks. This capability is crucial for addressing rare "long-tail" problems, as it enables the agent to perform risk assessment based on common sense, for instance, anticipating that a child might follow a rolling ball, even without having encountered the exact scenario in its training data. Building such models requires data at a much larger scale and necessitates a balance between the general knowledge of the VLM and the precision required in driving planning tasks, presenting a key research challenge and potential.

Agent Systems. The pursuit of robust, generalizable autonomous driving is increasingly drawing inspiration from human cognition. A human driver does not compute raw motor commands directly from visual input. Instead, the brain functions as a hierarchical system, where a conscious, reasoning mind sets high-level goals (e.g., "safely navigate this intersection"), while delegating the precise physical execution to well-honed motor skills. Mirroring this cognitive architecture offers a compelling path to overcome the brittleness of purely monolithic models. Following this principle, a prominent future trend indicates a paradigm shift from monolithic, black-box networks toward modular agent systems orchestrated by an LLM as the central cognitive engine. In this framework, rather than directly outputting low-level control signals, the LLM operates at a higher level of abstraction, performing reasoning, planning, and task decomposition. It then invokes specialized expert models or "tools," such as motion planners and controllers, for precise execution. This hierarchical architecture synergizes the powerful logical and causal reasoning of LLMs with the stability and precision of conventional modules. This synergy promises significant enhancements in interpretability, robustness, and the capacity to handle complex tasks.

World Models A World Model internally simulates or imagines the most probable evolution of the environment over a future horizon, conditioned on the current state and a hypothetical action from the agent. This internal simulation capability enables the agent to conduct extensive policy exploration and learning within a safe, zero-cost virtual space. Furthermore, the concept of a World Model can be framed as a self-supervised task to enhance the model’s understanding of the environment [56, 71]. By primarily relying on self-supervised learning, this approach significantly reduces the dependency on expensive, manually annotated data, opening up the possibility of leveraging vast amounts of unlabeled driving videos to drive the model’s evolution.

Cross-Modal Fusion. Although current VLA models have achieved significant progress in integrating semantic understanding, they are still largely confined to RGB visual inputs, which limits their precise perception of the 3D structure of the physical world. In contrast to camera data, depth image or LiDAR provides accurate, illumination-invariant 3D geometric information, which can directly compensate

for the inherent shortcomings of vision-only approaches in spatial relationship assessment. Therefore, developing next-generation VLA models that can effectively fuse the rich semantic information from cameras with the precise geometric data is critical to enhancing the decision-making robustness of end-to-end systems in complex 3D scenes.

Data Engine. The continuous evolution of end-to-end autonomous driving models is fundamentally contingent upon an efficient, automated, large-scale data engine. The core paradigm for such systems has shifted from an early data-driven model, focused merely on data volume, to a problem-driven approach aimed at resolving model deficiencies. As illustrated in Sec. 6.1, modern data engines are designed to precisely and automatically mine massive streams of real-world data to identify a vast number of high-value corner hard cases, *i.e.*, scenarios that trigger model failures or decision-making uncertainties. This methodology provides the large-scale, high-quality data essential for propelling the advancement of next-generation technologies.

8 CONCLUSION

This survey presents a systematic review of three end-to-end autonomous driving paradigms from a unified perspective, examining their respective datasets, model architectures, and learning strategies. Although current systems exhibit notable autonomous driving and generalization capabilities, they face persistent challenges, including the long-tail problem in data, ensuring system interpretability and safety, and the complexities of real-world deployment. Looking ahead, foundational models, data engines, and cross-modal integration represent promising research directions. Furthermore, reinforcement learning, agent systems, and world models offer potential pathways to achieving more advanced levels of autonomous driving. We advocate for the establishment of standardized assessment benchmarks and protocols to enable fair comparisons across diverse methodologies and to mitigate algorithm-specific data dependencies. Ultimately, we hope this comprehensive review offers the community novel insights and catalyzes future high-impact research in this domain.

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